WELCOME TO OMNI ACADEMY

Transforming People and Business | www.omni-academy.com



Course Title : AWS Cloud Watch

Trainer Name: Sohaib Ur Rehman

Amazon - CloudWatch

What is Cloud Watch?

"Amazon CloudWatch is a monitoring service for AWS cloud resources and the applications run on AWS. It is used to collect and track metrics, collect and monitor log files, and set alarms."





Amazon - CloudWatch

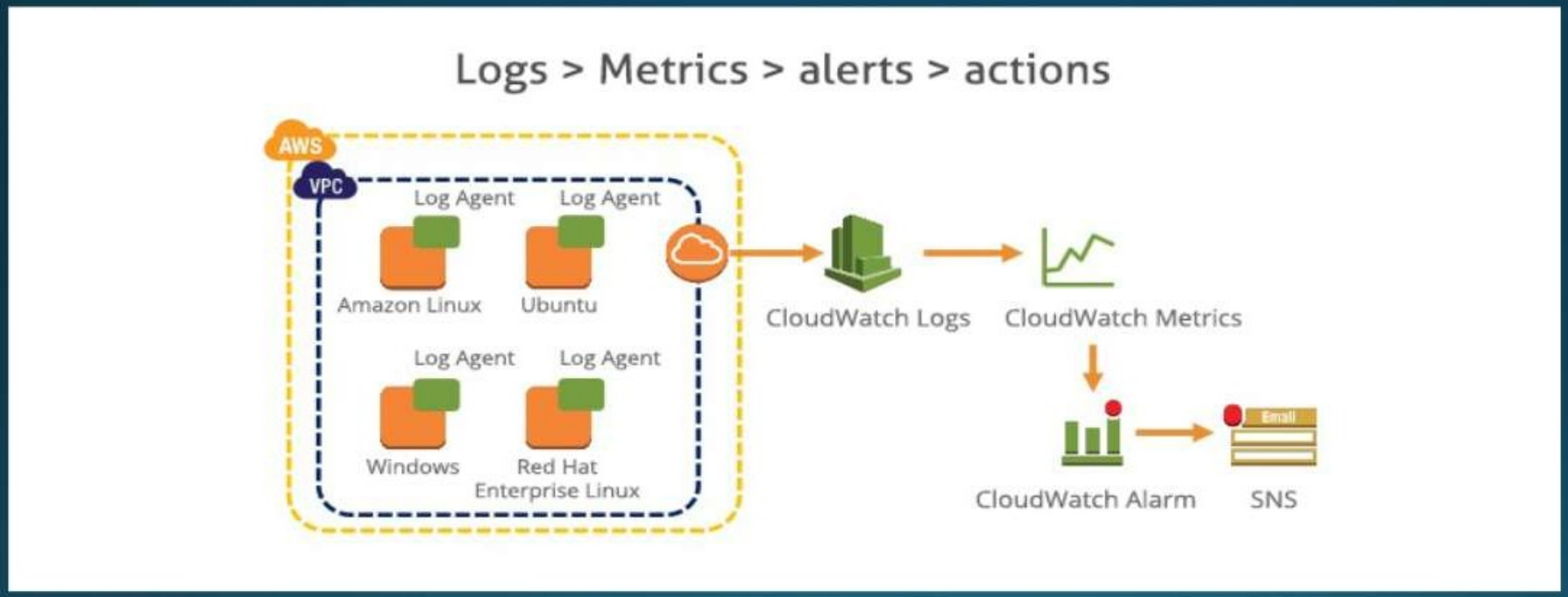
How Amazon CloudWatch works?

- Amazon CloudWatch is a metrics repository.
- An AWS service—such as Amazon EC2—puts metrics into the repository, and you retrieve statistics based on those metrics.
- If you put your own custom metrics into the repository, you can retrieve statistics on these metrics as well.





Amazon - CloudWatch





Amazon - CloudWatch

Dashboard

Amazon CloudWatch dashboards are customizable home pages in the CloudWatch console that you can use to monitor your resources in a single view, even those resources that are spread across different regions.

With dashboards, you can create the following:

- A single view for selected metrics and alarms to help you assess the health of your resources
- A common view of critical resource and application measurements that can be shared by team members







Amazon - CloudWatch

Amazon CloudWatch Prerequisite

Before getting started with Amazon CloudWatch it's important to know the items it enables th
user to monitor:

1- Amazon EC2 instances
2- Auto Scaling groups
3- EMR job flows
4- Estimated AWS charges
5- Amazon EBS volumes
6- Amazon SQS queues
7- Storage Gateway
8- Amazon RDS database instances in real-time
9- Elastic Load balancers
10- SNS topics
11- DynamoDB tables







Amazon - CloudWatch

CloudWatch is nothing but a repository of data.

t can receive data from two sources namely AWS Products and also the Custom Data.

There are scripts/agent available which we can configure in any machine in such a way that it would capture current data and send it to CloudWatch.

Amazon Charges for CloudWatch

t would charge per metrics. In EC2 there are 7 metrics. If it is basic monitoring it would be fre of cost. If it is for detailed monitoring, it would charge $2.10 per instance per month.







Amazon - CloudWatch

AMAZON CLOUDWATCH FEATURES	AMAZON CLOUDWATCH USAGE
1- Up-to-minute statistics 2- View graphs 3- Set alarms for your metric data 4- Use Auto Scaling to add/remove resources based on CloudWatch Metrics	1- Visibility into resource utilization 2- Operational performance 3- Overall demand patterns, including metrics such as CPU utilization, disk reads and writes, and network traffic.





Amazon - CloudWatch

Amazon Cloudwatch Monitoring

The Cloudwatch Alarm would be use to take 3 actions

1- Autoscaling

2- SNS (Simple Notification Service) – notifies through Email/SMS/http endpoint.

3- EC2 Action

4- System Manager Ops Center



Amazon - CloudWatch

Amazon Cloudwatch Metrics:

Namespaces:
It is a grouping to know what this metric belongs to. For example: AWS/EC2, AWS/AutoScaling AWS/ELB.

Dimensions:
Dimension is a name/value pair that you uniquely identify a metric. For example: AutoScaling GroupName, ImageId, InstanceID, InstanceType, Volume ID.

Timestamps:
To know what timestamp it had captured.

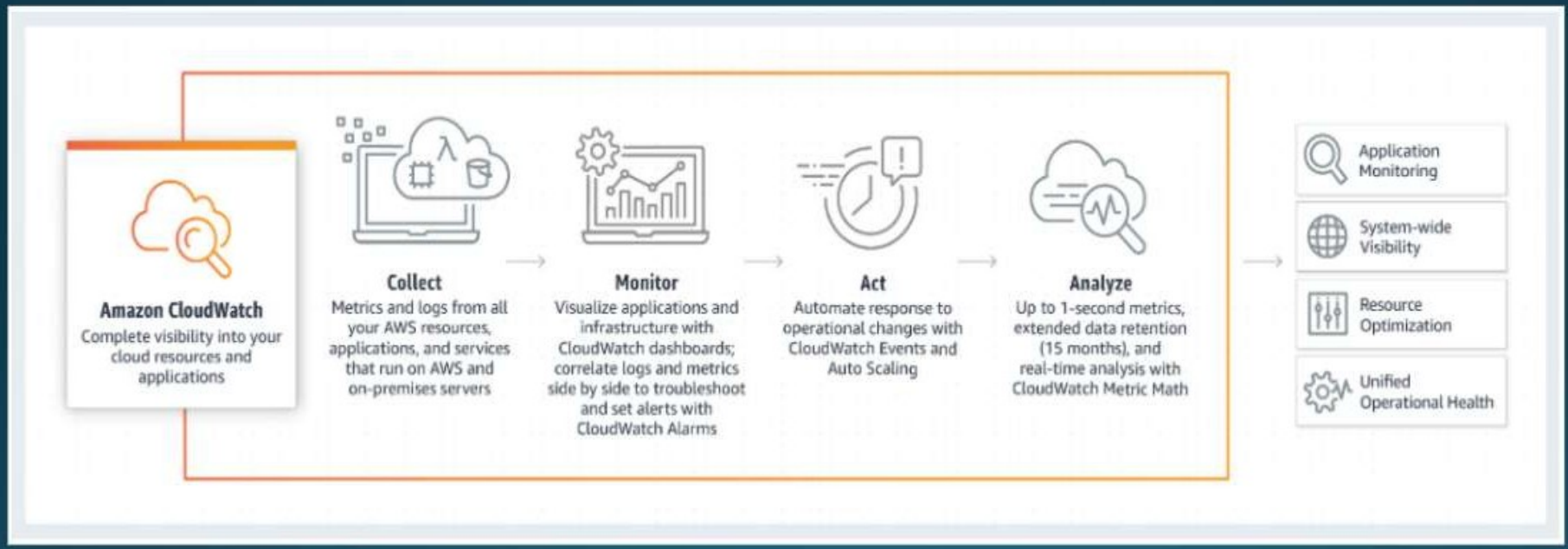
Units:
Unit represents the statistical unit of measure. For example: EC2 Networking metric in bytes.





Amazon - CloudWatch

How it works





Amazon - CloudWatch

Hands-On Lab

- Create a billing alarm which should be <= Current billing cost
- Create Dashboard
- Verify alarm notification via SMS & email.



